Model Predictive Control for Soil Moisture Regulation in Crops

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***Abstract*—This paper presents the design and implementation of a Model Predictive Controller (MPC) for regulating soil moisture in the root zone of a crop. A simplified dynamic model based on the soil water balance is employed, incorporating crop evapotranspiration and precipitation as external disturbances, as well as a moisture-dependent deep drainage term. The controller is formulated as a non-linear optimization problem with physical and operational constraints, solved using numerical optimization tools. Additionally, a preliminary Grey-Box approach using a simple neural approximation is explored, aimed at evaluating its potential future integration as a complement to the physical model. Simulation results demonstrate that the MPC is capable of maintaining soil moisture close to a reference value, anticipating climatic disturbances and avoiding over-saturation states.**

***Index Terms*—water balance, CasADI, predictive control, deep drainage, evapotranspiration, automatic irrigation.**

I. Introduction

Efficient water management in agriculture is a key challenge in the current context of water scarcity, climate variability, and increasing food demand. Water availability in the root zone directly influences crop growth and yield; both deficit and excess moisture generate adverse effects on the soil–plant system, such as water stress, leaching, and waterlogging. Therefore, precise regulation of soil moisture has become a central objective of modern irrigation systems [1].

Traditional irrigation schemes are primarily based on open-loop control, using fixed calendars or empirical rules [2]. Although widely used, these methods do not explicitly consider the dynamics of the soil–plant–atmosphere system nor the influence of variable climatic disturbances, such as precipitation and evapotranspiration, which can lead to inefficient use of water resources [3]. The FAO-56 standard provides a recognized framework for estimating crop water demand [4], but its practical application often lacks dynamic feedback of the soil state.

As an alternative, closed-loop control strategies have been proposed, such as PID controllers and moisture threshold-based methods [5]. However, these techniques present limitations in anticipating future disturbances and handling physical constraints in non-linear systems with delays [9]. In this context, Model Predictive Control (MPC) has consolidated as a suitable tool for soil moisture regulation, allowing the explicit incorporation of dynamic models, physical constraints, and climatic forecasts within an optimization problem [6]–[8].

This work proposes a predictive control scheme based on a dynamic model of the root zone water balance, which explicitly incorporates crop evapotranspiration, precipitation as an external disturbance, and a moisture-dependent deep drainage term. Likewise, a preliminary Grey-Box approach is explored using a simple neural network to capture unmodeled residual dynamics, maintaining the interpretability of the physical model. The article is organized as follows: Section II presents the dynamic system model; Section III describes the formulation and implementation of the predictive controller; and Section IV analyzes the simulation results.

Additionally, a Grey-Box approximation using a simple neural network structure is preliminarily explored, oriented towards capturing residual dynamics not explicitly represented by the physical model. This approximation is considered a potential complement to the base model, preserving its interpretability and laying the groundwork for future developments.

The rest of the article is organized as follows: Section II presents the dynamic model of the system based on the soil water balance; Section III describes the formulation of the predictive controller and its numerical implementation; and Section IV analyzes simulation results under different scenarios.

II. Dynamic System Model

This section presents the dynamic model of the soil–plant–atmosphere system that describes the evolution of soil moisture based on the main water flows: irrigation, precipitation, evapotranspiration, and drainage. This model constitutes the mathematical basis for the design of the model predictive controller.

A. Soil water balance

The system is modeled through a water balance equation in the root zone, where the state variable is the volumetric soil moisture θ. The continuous dynamics of the system are expressed as:

$$\frac{d\theta}{dt} = \frac{1}{Z_r}\left(I(t) + P(t) - ET_c(t,\theta) - D(t,\theta)\right) \tag{1}$$

where θ is the volumetric soil moisture, Z_r is the effective depth of the root zone (mm), $I(t)$ and $P(t)$ represent irrigation and precipitation, $ET_c(t,\theta)$ is the actual crop evapotranspiration, and $D(\theta)$ is the deep drainage, following the classical water balance approach proposed in [4].

B. Precipitation modeling

In the early stages, precipitation was modeled as a rectangular pulse using step functions:

$$P(t) = P_0[H(t - t_1) - H(t - t_2)] \tag{2}$$

where $P(t)$ is the precipitation in mm/day, P_0 is the intensity of the rain event, t_1 and t_2 delimit the start and end of the event, and $H(\cdot)$ is the Heaviside function.

This formulation introduces discontinuities in time, which does not adequately represent the actual evolution of rainfall. For this reason, a continuous formulation based on a double sigmoid was subsequently adopted:

$$P(t) = P_{\max}\left[\frac{1}{1 + e^{-a(t-t_1)}} - \frac{1}{1 + e^{-a(t-t_2)}}\right] \tag{3}$$

where $P_{\max}$ is the maximum rain intensity, $a > 0$ is the smoothing parameter that controls the growth and decay slope of the event, and t_1, t_2 represent the beginning and end of the event.

This function describes a progressive increase in rain, followed by a gradual decrease, reproducing more realistically the dynamics of a precipitation event.

C. Reformulation of crop evapotranspiration

In a first approximation, crop evapotranspiration can be considered as an external signal dependent only on time. However, this formulation does not reflect the physical fact that the actual crop evapotranspiration is reduced when the soil presents a water deficit.

In order to explicitly introduce this effect, evapotranspiration is reformulated as:

$$ET_c(t, \theta) = ET_{c,\text{pot}}(t)\, K_s(\theta) \tag{4}$$

where $ET_{c,\text{pot}}(t)$ is the potential crop evapotranspiration, determined from climatic models, and $K_s(\theta)$ is the water stress coefficient, defined as a non-linear and bounded function $K_s : [0, 1] \to [0, 1]$, according to FAO-56 [4].

The coefficient $K_s(\theta)$ models the progressive reduction of evapotranspiration when soil moisture falls below a critical threshold, taking values close to one in optimal conditions and approaching zero under water stress.

Two critical moisture levels are considered: the permanent wilting point θ_{wp} and the field capacity θ_{fc}, which delimit the soil's hydraulic operating range.

Initially, water stress was modeled in a binary form:

$$K_s(\theta) = \begin{cases} 0, & \theta \leq \theta_{wp} \\ 1, & \theta \geq \theta_{fc} \end{cases} \tag{5}$$

This formulation introduces abrupt changes that do not adequately represent the physiological behavior of the plant and also generate discontinuities that hinder MPC optimization.

For this reason, K_s was reformulated as a continuous function:

$$K_s(\theta) = \begin{cases} 0, & \theta \leq \theta_{wp} \\ \frac{\theta - \theta_{wp}}{\theta_{fc} - \theta_{wp}}, & \theta_{wp} < \theta < \theta_{fc} \\ 1, & \theta \geq \theta_{fc} \end{cases} \tag{6}$$

The function models the progressive increase in water stress as a function of soil moisture, using continuous formulations that avoid abrupt saturations and guarantee differentiability, a key requirement for the MPC.

In this context, $ET_{c,\text{pot}}$ acts as an external climatic disturbance, while $K_s(\theta)$ introduces non-linear feedback that relates the hydraulic state of the soil with the actual transpiration of the crop, allowing the separation of atmospheric demand from the soil–crop response.

D. Deep drainage modeling

Deep drainage represents water losses when soil moisture exceeds field capacity, causing flow into deeper layers. This phenomenon depends solely on the water state and is not controllable.

In the model, drainage is incorporated into the system dynamics without being defined as a state or control variable. Its activation upon exceeding field capacity allows for a physically consistent representation of moisture losses associated with over-saturation.

In this work, drainage is initially modeled as a non-linear function of soil moisture:

$$D(\theta) = \begin{cases} 0, & \theta \leq \theta_{fc} \\ k_d(\theta - \theta_{fc}), & \theta > \theta_{fc} \end{cases} \tag{7}$$

where k_d is a positive coefficient representing the drainage rate. This approach, common in modeling gravitational losses [17], activates drainage only when $\theta > \theta_{fc}$. However, this instantaneous activation is not physically realistic, as percolation increases gradually as soil pores become saturated.

To resolve this issue, drainage was reformulated using a softplus-type smooth function:

$$D(\theta) = K_d \frac{1}{\alpha} \ln\left(1 + e^{\alpha(\theta - \theta_{fc})}\right). \tag{8}$$

This function preserves two key properties:

1) For $\theta \ll \theta_{fc}$, drainage is practically zero.
2) For $\theta > \theta_{fc}$, drainage grows approximately linearly with θ, as in the original piecewise model.

The parameter α controls the speed of the transition. Thus, the smoothed model maintains the physical coherence of the percolation process but eliminates the discontinuities that affect MPC performance.

III. Control Methodology

This section describes the design of the proposed predictive controller, whose objective is to regulate soil moisture θ around a reference value, simultaneously optimizing the use of water resources.

The general structure of the system is shown in Figure 1, where the MPC uses the dynamic model presented in Section

II to predict the future evolution of the system and calculate the optimal irrigation action $I(t)$, explicitly considering climatic disturbances.

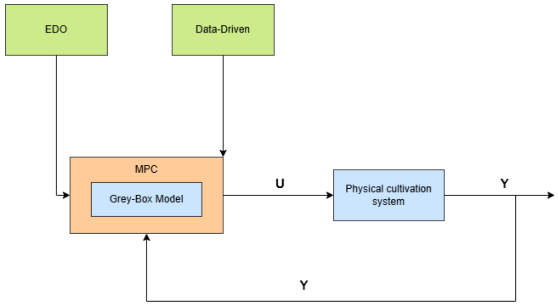

Figure 1. Block diagram of the irrigation control system.

A. Controller variables

To implement the MPC, the water balance is discretized using the forward Euler method with a daily time step (dt=1). This discrete-time formulation allows modeling the recursive evolution of the volumetric soil moisture $\theta(k)$ as a function of the applied irrigation $I(k)$ and climatic disturbances, facilitating efficient computational optimization.

B. Cost function

The objective function penalizes both the deviation of moisture from a reference value θ_{ref} and the excessive use of irrigation:

$$J = \sum_{k=1}^{N} (\theta(k) - \theta_{\text{ref}})^2 + \lambda I(k)^2 \tag{9}$$

where N is the prediction horizon, θ_{ref} is the desired reference moisture, and λ is a weighting parameter that penalizes excessive use of irrigation, following standard MPC formulations [6].

C. Constraints

The MPC incorporates physical constraints on the state and control action to guarantee viable trajectories. Soil moisture is bounded as $\theta_{min} \leq \theta(k) \leq \theta_{max}$, with $\theta_{min} = 0.1$ and $\theta_{max} = 0.4$, while the irrigation action satisfies $0 \leq I(k) \leq I_{max}$, where I_{max} represents the maximum capacity of the system.

These constraints are imposed throughout the prediction horizon, allowing the controller to anticipate their fulfillment. The inclusion of integral action compensates for persistent errors without generating saturation of the irrigation signal.

D. Grey-Box Model

Grey-Box modeling combines physical knowledge (*White-Box*) with data-based flexibility (*Black-Box*). In this work, a residual structure is used, where the physical model defines the global trend of the system and an Artificial Neural Network (ANN) estimates the residual term $r(k)$. The ANN is of the *feed-forward* type, with hidden layers and non-linear activation functions (*tanh*), and is trained to minimize the squared error between the response of the actual plant and that of the simplified model.

IV. Simulation Results

The MPC controller was implemented in MATLAB using the CasADi library and evaluated through numerical simulations in a controlled computational environment. The complete MATLAB–CasADi implementation is available in a public repository.[1] The simulations were developed progressively to analyze the controller's performance under different levels of dynamic model complexity.

Initially, a base model without disturbances was considered, to which precipitation, state-dependent evapotranspiration, non-linear deep drainage, smoothing functions, physical saturations, integral action, and variable moisture references were added. These scenarios allowed for the evaluation of MPC behavior under dynamic climatic conditions.

The climatic signals used were synthetic, facilitating the individual analysis of each component's effect. Finally, the scheme was extended to a Grey-Box approach by incorporating a residual term estimated by a neural network.

A. Base model without disturbances

In the first stage, the MPC was evaluated on a simplified hydrological model, considering only soil moisture and irrigation without external disturbances. Figure 2 shows that the controller regulates moisture stably toward the reference, generating a smooth irrigation action and confirming the correct formulation of both the model and the control scheme.

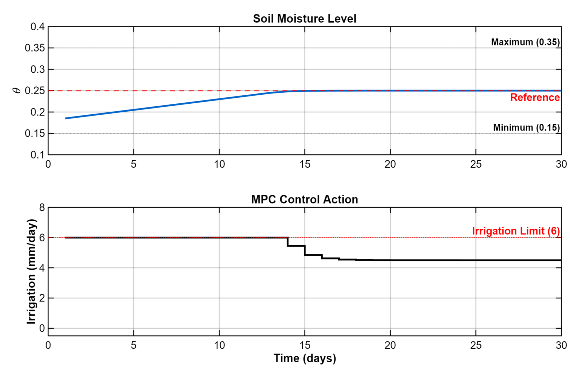

Figure 2. Evolution of soil moisture and irrigation signal for the base model without disturbances.

B. Inclusion of precipitation as an external disturbance

In this stage, precipitation was incorporated as a known external disturbance directly affecting the soil water balance. Figure 3 shows that the MPC reduces or cancels irrigation during rain events, anticipating the additional water input and maintaining moisture close to the reference.

[1] https://github.com/MiguelPanqueva2023/soil-moisture-mpc-greybox

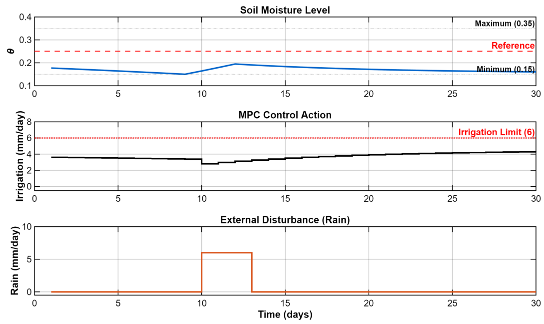

Figure 3. Comparison of the irrigation signal in the presence and absence of precipitation.

C. Time-varying crop evapotranspiration

In this stage, crop evapotranspiration was modeled as a time-varying signal, representing daily climatic fluctuations and acting as an exogenous disturbance. Figure 4 shows that the MPC compensates for these variations through coherent adjustments in the irrigation signal, keeping soil moisture close to the reference value.

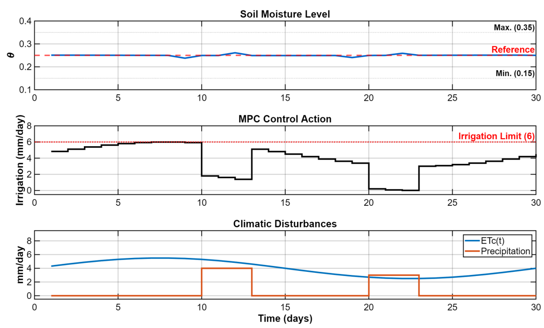

Figure 4. Time-varying crop evapotranspiration and irrigation response obtained via MPC.

D. Moisture-dependent crop evapotranspiration

In this stage, the dependence of evapotranspiration on soil moisture was incorporated through a water stress coefficient, introducing a non-linearity that reduces evapotranspiration under water deficit conditions. Figure 5 shows that this formulation attenuates excessive moisture drops and allows the MPC to maintain stable and physically coherent regulation.

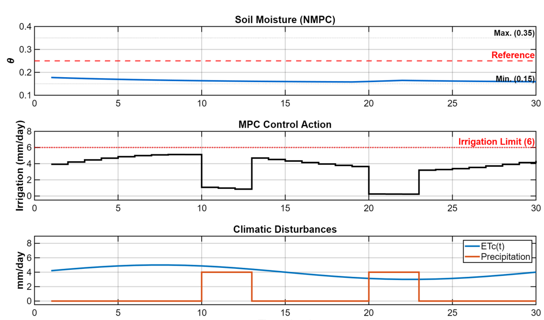

Figure 5. Water stress coefficient and soil moisture evolution.

E. Inclusion of non-linear deep drainage

Next, deep drainage was incorporated as a non-linear function dependent on soil moisture, activating only when field capacity is exceeded and representing unavoidable physical losses. Figure 5 (referenced as drainage mechanism) shows that this mechanism prevents excessive water accumulation, while the MPC compensates for such losses by adjusting irrigation to keep moisture near the reference.

F. Introduction of smoothing in non-linear processes

In this stage, the discontinuous functions for effective evapotranspiration and deep drainage were replaced by smooth, differentiable formulations, eliminating abrupt changes in the system dynamics. As shown in Figure 6, this reformulation generates continuous and physically coherent soil moisture trajectories.

As a result, the MPC adjusts irrigation progressively, avoiding oscillations and improving numerical stability and the physical consistency of the system.

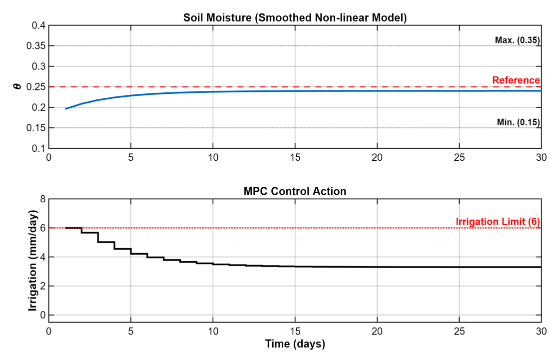

Figure 6. MPC response under smooth physical saturations in soil moisture and irrigation.

G. Incorporation of smooth physical saturations and integral action

Smooth saturations and integral action were incorporated into the MPC, ensuring realistic operation and improving robustness against disturbances.

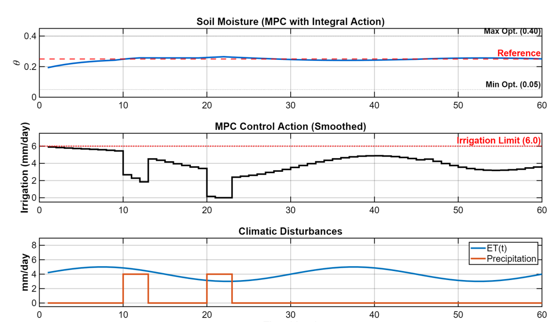

Figure 7. Effect of smooth physical saturations and integral action on soil moisture stability.

H. Substitution of the rainfall model with a double sigmoid function

Figure 8 shows that the use of continuous sigmoidal precipitation allows the MPC to adjust irrigation in an early and stable manner, avoiding abrupt actions.

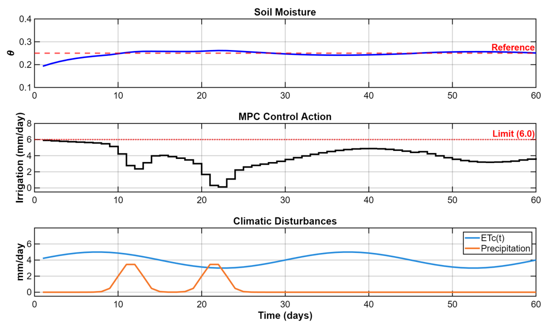

Figure 8. Continuous modeling of rain events using a double sigmoid function.

I. Variable moisture reference and climatic scenarios

The moisture reference and climatic variables are defined as time-dependent functions to evaluate the controller's performance under realistic dynamic conditions.

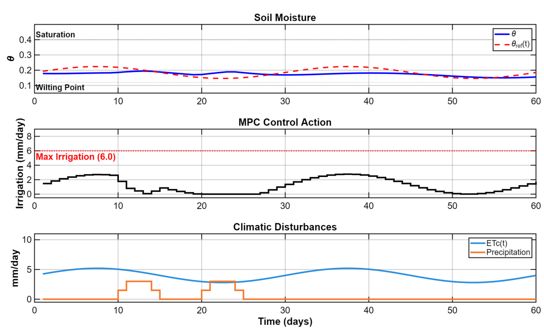

Figure 9. MPC performance under variable moisture references and contrasting climatic scenarios.

Figure 9 shows that the MPC follows variable references and adjusts irrigation to external disturbances, keeping soil moisture within optimal ranges in dynamic scenarios.

J. Synthetic data generation and residual analysis

Synthetic data were generated by integrating the predictive controller with the physical model of the soil water balance under realistic operating conditions. The controller calculates the optimal irrigation action using only the physical model, while the simulated plant includes smooth disturbances representing unmodeled dynamics and bounded noise.

The residual is defined as the difference between the model response and the perturbed plant response, stored alongside states, control action, and climatic disturbances. This process allows for the construction of a structured dataset that preserves the dominant physical dynamics and facilitates learning residual effects through a Grey-Box approach.

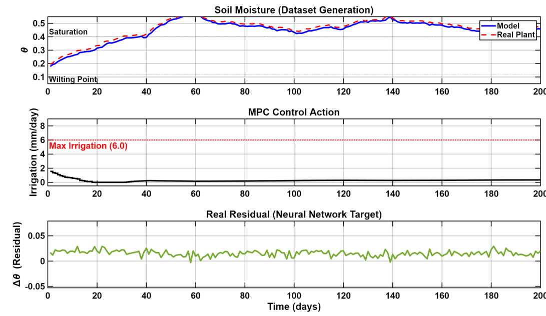

Figure 10. Synthetic data generation for the Grey-Box model: (top) soil moisture evolution obtained with the physical model and the actual simulated plant; (center) irrigation signal calculated by the MPC; (bottom) residual term defined as the target for neural network training.

Figure 10 demonstrates that the physical model reproduces the main soil moisture dynamics, while the simulated plant shows smooth deviations due to unmodeled effects. The irrigation signal generated by the MPC is stable and converges to near-constant values upon reaching the reference.

The residual is bounded and exhibits temporal structure, indicating that it corresponds to learnable dynamics rather than random noise, thus validating the Grey-Box approach based on residual compensation.

K. Training of the residual neural network

A residual neural network was trained with synthetic data to approximate the error between the physical model and the simulated plant under a Grey-Box approach. The network, trained by minimizing the mean squared error, captures the main trend of the residual without affecting MPC stability (Figure 11).

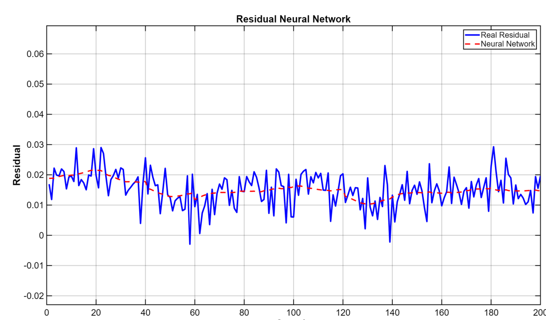

Figure 11. Comparison between the system's actual residual and the residual estimated by the trained neural network.

In this stage, the neural network is validated in both an offline identification scheme and a preliminary integration within the MPC. However, the formal analysis of closed-loop stability and validation with real data are left for future work.

L. Integration of the Grey-Box model into the MPC controller

The residual neural network was integrated into the MPC under a Grey-Box approach, where the physical model defines the main dynamics and the network corrects unmodeled errors.

This compensation improves reference tracking without significantly increasing the control action, as shown in Figure 12.

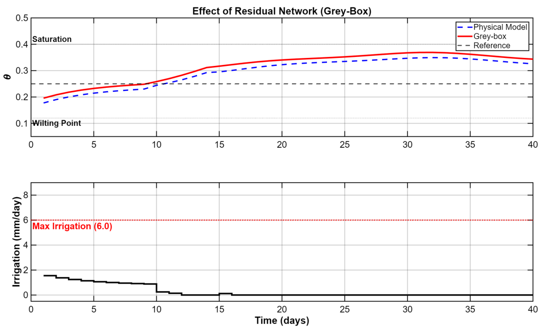

Figure 12. Comparison between the physical model and the Grey-Box scheme within the MPC.

V. Discussion

The results show that the gradual incorporation of physical complexity improves the representation of soil dynamics and MPC performance by introducing physically coherent nonlinearities such as water stress and state-dependent drainage.

The use of smoothing functions and soft physical constraints is fundamental to guaranteeing numerical stability and controller robustness against climatic disturbances. Furthermore, the Grey-Box approach allows for increased model precision while maintaining interpretability, by limiting neural learning to residual dynamics.

VI. Conclusions and Future Work

An MPC controller based on a dynamic soil water balance model was developed for soil moisture regulation in crops, progressively incorporating relevant physical phenomena. Simulation results show stable and robust behavior under different operating scenarios.

The inclusion of realistic physical constraints and a residual Grey-Box approach allowed for an improved representation of the system dynamics without compromising interpretability. Future work will involve validation with real data and analysis of controller stability and robustness against modeling errors and uncertainties.

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